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Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

ENSO El Niño–Southern Oscillation

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

PC Principal Component

PDO Pacific Decadal Oscillation

VC Vector of Coefficients

Theoretical Foundations

1.1 | Principal Component Analysis

Principal Component Analysis (PCA) is a technique that transforms a large set of variables into a smaller one that still contains most of the information in the original set by identifying the directions along which the data varies the most. As we will see, this is achieved by projecting the data onto the eigenvectors of the covariance matrix of the data, where these eigenvectors, or vector of coefficients, correspond to the directions of maximum variance, and the associated eigenvalues represent the magnitude of variance in these directions. This can be extremely helpful when dealing with high-dimensional data where the goal is dimensionality reduction, data compression, feature extraction or visualization of these high-dimensional data.

The first description of PCA can be traced back to Pearson [45] in 1901 and Hotelling [30] in 1936, but it was not until 1956 when Lorenz [40] popularized PCA in the field of meteorology under the term *empirical orthogoanal function analysis*¹ Subsequently, PCA has been widely used in meteorology and climate science to identify the dominant modes of variability in the atmosphere and oceans with entire chapters of popular textbooks being dedicated to this technique [e.g. 24, 46, 52].

In the following we will quickly derive PCA from the maximum variance formulation², [e.g. 3, pp. 561-563], in which the data is orthogonally projected onto a lower dimensional linear space, known as the principal subspace, such that the variance of the projected data is maximized. Then we will discuss the main shortcomings of PCA in particular in the context of climate science and some of the most common ways to overcome these.

nicrie: make link to Karhunen-Loève expansion

¹Indeed, until today EOF analysis seems to be the preferred term used in meteorology and climate science. We will stick to the more commonly used term PCA though.

²This is what Hotelling [30] used to derive PCA. An alternative derivation is the minimum error formulation by Pearson [45], in which we seek a linear projection that minimizes the average projection cost, defined as the mean squared distance between the data points.

1.1.1 Derivation based on Maximum Variance Formulation

Consider a data set represented as a two dimensional $N \times D$ matrix $\mathbf{X}$ of N observations with D variables such that the element x_{ni} represents the n^{th} observations of the i^{th} variable. For simplicity, we assume that the data is centered, i.e. the mean of each variable is zero, $\bar{x}_i = 0$ for all $i = 1, \dots, D$.

Our goal is to project the data onto a space having dimensionality $M < D$ while maximizing the variance of the projected data. To begin with, consider the projection onto a one-dimensional space ($M = 1$). We can define the direction of this space using a D -dimensional vector $\mathbf{v}_1$, which for convenience we shall choose to be a unit vector so that $\mathbf{v}_1^T \mathbf{v}_1 = 1$. Note that we are only interested in the direction of $\mathbf{v}_1$, not its magnitude. Each data point $\mathbf{x}_n$ is then projected onto a scalar value $\mathbf{v}_1^T \mathbf{x}_n$, so that the variance of the projected data is given by

$$\frac{1}{N-1} \sum_{n=1}^{N-1} \{\mathbf{v}_1^T \mathbf{x}_n - \mathbf{v}_1^T \bar{\mathbf{x}}\}^2 = \frac{1}{N-1} \sum_{n=1}^{N-1} \{\mathbf{v}_1^T \mathbf{x}_n\}^2 = \mathbf{v}_1^T \mathbf{S} \mathbf{v}_1, \quad (1.1)$$

where $\mathbf{S}$ is the covariance matrix defined by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^T = \frac{1}{N-1} \mathbf{X}^T \mathbf{X}. \quad (1.2)$$

Then, the problem can be formulated as a optimization problem in which we seek

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^T \mathbf{v}_1 = 1. \quad (1.3)$$

To find a solution to Equation (1.3), we can maximize the projected variance $\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint from above, $\mathbf{v}_1^T \mathbf{v}_1 = 1$, we introduce a Lagrange multiplier that we shall denote by λ_1 , and then make an unconstrained maximization of

$$\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^T \mathbf{v}_1). \quad (1.4)$$

By setting the derivative with respect to $\mathbf{v}_1$ equal to zero, we see that this quantity will have a stationary point when

$$\mathbf{S}\mathbf{v}_1 = \lambda_1\mathbf{v}_1. \quad (1.5) \quad 1$$

which says that $\mathbf{v}_1$ must be an eigenvector of $\mathbf{S}$ and λ_1 the corresponding eigenvalue. If we left-multiply by $\mathbf{v}_1^\top$ and make use of the normalization condition, we see that the variance is given by 2
3
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$$\mathbf{v}_1^\top \mathbf{S}\mathbf{v}_1 = \lambda_1 \quad (1.6) \quad 5$$

and so the variance will be a maximum when we set $\mathbf{v}_1$ equal to the eigenvector having the largest eigenvalue λ_1 . The eigenvector is known as the first *vector of coefficients*³ (VC) of the data set. 6
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We can define additional VCs in an incremental fashion by choosing each new direction to be that which maximizes the projected variance amongst all possible directions orthongonal to those already considered. If we consider the general case of an M -dimensional projection space, the optimal liner projection for which the variance of the projected data is maximized is now defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the data covariance matrix $\mathbf{S}$ corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This is easily shown using proof by induction (Appendix A), yielding the consice expression 9
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$$\mathbf{S}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}, \quad (1.7) \quad 16$$

where by construction the eigenvector matrix $\mathbf{V}$ is orthogonal, i.e. $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_M$. 17

Thus we see how the eigenvectors of $\mathbf{S}$ arise naturally when we try to find a new coordiante system "along" regularities in the data. For what to do when the regularities in the data are not aligned in the orthogonal directions, see Section 1.1.2. 18
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From the spectral theorem for normal matrices [e.g. 29, p. 133] it follows that the symmetric covariance matrix $\mathbf{S}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding eigenvalues λ_i : 21
22
23

³Naming convention depends a lot on the scientific domain. For instance, the pair of eigenvectors (here $\mathbf{V}$) and corresponding data projections (here $\mathbf{P}$) are called the principal components and principal component scores in machine learning [e.g. 3], vector of coefficients/loadings and principal components in statistics [e.g. 32] or empirical orthogonal functions and principal components/expansion coefficients in climate science and meteorology [e.g. 52]. Here I try to be consistent and follow the convention used in statistics.

$$\mathbf{S} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^T. \quad (1.8)$$

Therefore, each eigenvalue λ_i represents the amount of variance described by each component which is typically expressed in form of a fraction γ_i for the corresponding i^{th} component:

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.9)$$

Projecting the data $\mathbf{X}$ onto the M dimensional subspace thus provides the data representation in the projection space, called the raw principal components (PCs):

$$\mathbf{P} = \mathbf{XV}, \quad (1.10)$$

which gives also the possibility to reconstruct the original data from (a subset of) the projected data:

$$\mathbf{X} = \mathbf{PV}^T. \quad (1.11)$$

However, computing the eigenvector decomposition is not the most efficient way computationally seen. A more efficient way to compute the eigenvectors involves the singular value composition (SVD) [e.g. 29, p. 150] of the data matrix

$$\mathbf{X} = \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^T, \quad (1.12)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are respectively $N \times M$ and $M \times D$ unitary matrices, i.e. $\mathbf{U}'^T \mathbf{U}' = \mathbf{V}'^T \mathbf{V}' = \mathbf{I}_M$ where $M \leq \min(N, D)$ is the rank of $\mathbf{X}$ and $\mathbf{I}_M$ is the identity matrix of order M . The matrix $\mathbf{\Sigma}$ is a diagonal matrix of the singular values σ_i of $\mathbf{X}$, i.e. $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M \geq 0$.

The covariance matrix $\mathbf{S}$ can be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X} = \frac{1}{N-1} \mathbf{V}' \mathbf{\Sigma}^T \mathbf{U}'^T \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^T = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T, \quad (1.13)$$

1.1 Principal Component Analysis

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where, by comparison with (1.7), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1} \mathbf{\Sigma}^T \mathbf{\Sigma}$. This last equation directly links the eigenvalues of the covariance matrix to the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.14)$$

Comparing (1.11) with (1.12) shows that the raw principal component scores $\mathbf{P}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P} \mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \mathbf{P} \mathbf{\Lambda}^{-1/2},$$

or alternatively $\mathbf{P} = (N-1)^{1/2} \mathbf{U} \mathbf{\Lambda}^{1/2}.$ (1.15)

so that $\mathbf{U}^T \mathbf{U} = \mathbf{I}_M$ and the principal component scores $\mathbf{U}$ are unitary, thus uncorrelated.

A common scaling for the vector of coefficients $\mathbf{V}$ is where they are scaled by the square root of the corresponding eigenvalues, which are commonly called the *loadings*, $\tilde{\mathbf{V}} = \mathbf{V} \mathbf{\Lambda}^{1/2}$. Note that the loadings $\tilde{\mathbf{V}}$ are not strictly unitary, although they are still a generalized unitary matrix since

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda} \quad (1.16)$$

is a diagonal matrix with non-negative real entries on the diagonal. With this scaling the PCA formulation Equation 1.11 becomes

$$\mathbf{X} = (N-1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^T. \quad (1.17)$$

nicrie: generlize the section to complex numbers, importantly orthogonal become unitray

Computational considerations Computationally, solving the eigendecomposition of the covariance matrix scales with $O(D^3)$ making it infeasible for modern data sizes in e.g. climate science where typical data sets⁴ are of the order $D = O(10^6)$, but can be even much larger in practice.

⁴For instance, a single climate variable of ERA5, the go-to standard reference for state-of-the-art reanalysis climate data is distributed on a regular 0.25° grid of Earth surface which translates to $D = 1\,036\,800$ grid cells.

In practice, alternative, more efficient algorithms involving the singular value decomposition (SVD) or the power methods [19] have been used instead from the beginning of the 21st century onward. More recently, randomized low-rank approximation techniques [15, 22] have allowed to considerably reduce the computational complexity to $O(ND \log(M) + (N + D)M^2)$ offering feasible alternatives for large, out-of-memory datasets (Table 1.1).

Shortcomings Despite its popularity and wide applicability, PCA has several limitations and potential drawbacks that should be considered when using this technique:

1. **Linearity assumption:** PCA assumes that the data's underlying structure is linear, which may not always hold true, especially in complex systems like climate data where nonlinear relationships can be significant.
2. **Interpretability of VCs:** In addition to the linearity assumption, the dominant modes of variability are represented in an orthogonal subspace spanned by the VCs. This orthogonality is a mathematical construction that may not always correspond to the physical reality of the data, potentially leading to misinterpretations of the results (see Section 1.1.2).
3. **Variance focus:** Capturing the directions of maximum variance may not always correspond to the most important features for a given application, potentially overlooking critical but less variable information.
4. **Static nature:** As a static method PCA does not account for temporal dynamics or changes over time, which can be particularly limiting in climate science where time evolution is a key aspect.
5. **Sensitivity to outliers:** PCA can be sensitive to outliers, as extreme values can disproportionately influence the VCs, leading to misleading results.
6. **Sensitivity to scaling:** The results are sensitive to the scaling of the input data, meaning that variables with larger ranges can dominate the VCs unless the data is appropriately standardized.
7. **Loss of information:** Not a shortcoming by itself because it's sometime the goal of PCA, but it should be taken into account that by reducing dimensionality, PCA inherently discards some information, which might lead to the loss of important details that could be crucial for accurate analysis and interpretation in specific applications.

Over the last decades a plethora of modifications and extensions have been proposed to overcome these limitations. We only mention a few here that are particularly relevant in the context of climate science. These include various forms of kernel PCA [8, 51, 56] to address the linearity assumption, rotated [49] and sparse [23] PCA for interpretability, robust PCA [11] to address the sensitivity to outliers, and extended [6, 55] and complex [1, 27] PCA to

Table 1.1 Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [3] and Halko *et al.* [22].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

account for temporal dynamics. Although not strictly a form of PCA, eigenmethods like optimal persistence analysis (OPA) [13, 14], which allows to filter for spatial patterns linked to increased persistence, and Principal Oscillation Pattern (POP) analysis [25] that provides a mean to analyze dynamical systems, are an example of related approaches with the goal to gain insights about the dominant pattern of a physical field in climate science.

In the following sections, we will discuss some of the shortcomings and proposed solutions in more detail: in particular we will discuss how an additional rotation of the PCA solution can help address the challenge of physical interpretability (Section 1.1.2) and how complexifying the data provides a means to better understand time evolution patterns (Section 1.1.3).

1.1.2 Sparse PCA via Rotation

Thanks to its maximum-variance-explained property and its orthogonality property which makes it optimal in a least-square sense [e.g. 3], PCA is indeed an ideal tool for dimensionality reduction, to identify principal axes along which the covariance among all variables is distributed, or to examine interrelationships or covariation between stations, grid-points, or variables. However, the use of PCA to locate the causes for the variation or to derive physical modes of variability is ill-advised [32, 37, 49].

For one, there is no reasons why real-world processes should follow the artificial orthogonality property of the VCs imposed by PCA. Second, the dense nature of the VCs – that is all elements of the eigenvectors are non-zero – makes it often very difficult to make sense out of it.

In the context of climate and environmental science, where variables are often represented across space (such as a longitude-latitude grid), these properties can under certain conditions [7, 36, 49] promote the creation of recognizable series of spatial patterns in the VCs – known as the Buell patterns [7, 49] – and which often consist of a positive pattern for the first component, subsequently followed by increasing polarity i.e. bipole, tripole, quadrupole etc. for the subsequent components (Figure 1.1). These and other drawbacks often prevent the standard PCA to be related with the underlying physical processes [cf. e.g. 49] which have led to the

promotion of alternative approaches where the general idea is to seek a PCA solution whose VCs are sparse, i.e. have many zero or close-to-zero elements, for which, as a side effect, either the orthogonality property of the VCs is lost, the PCs are not correlated anymore or both.

Traditionally, the most common approach in climate science to obtain a sparse PCA solution has been achieved via rotation techniques [28, 48, 49], an approach borrowed from factor analysis [cf. 32, pp. 269]. In this "factor analysis viewpoint" of PCA [32, pp. 150-161], a linear transformation is applied to a subset⁵ of the VCs to obtain a new set of rotated VCs that are "simpler" in some heuristically defined way, which are hoped to align better with the underlying physical processes.

To understand this, recall Equation (1.17) to see that as soon as we give up the orthogonality constrain (either for the VCs or the PCs), the PCA formulation exhibits a inherent indeterminacy such that any arbitrary non-singular transformation $\mathbf{R}$ still satisfies the PCA formulation [cf. 49, p. 305]:

$$\mathbf{X} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^T = (\mathbf{N} - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^T. \quad (1.18)$$

A rotated PCA solution will exploit this by defining the transformation $\mathbf{R}$ in a way such that the rotated loadings⁶ $\tilde{\mathbf{V}}_{\mathbf{R}} = \tilde{\mathbf{V}} \mathbf{R}$ maximize a simplicity criterion f ,

$$\underset{\mathbf{R}}{\operatorname{argmax}} f(\tilde{\mathbf{V}}_{\mathbf{R}}). \quad (1.19)$$

Typically, these linear transformations are categorized into two classes: *orthogonal*⁷ rotation and *oblique* rotation. Orthogonal rotation techniques are defined as rotation where $\mathbf{R}$ is an orthogonal matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}_M$, while for oblique rotation $\mathbf{R}$ is a general non-singular matrix. The most important consequence of this distinction is that the rotated PCs remain uncorrelated for orthogonal rotation, since $\mathbf{U}_{\mathbf{R}}^H \mathbf{U}_{\mathbf{R}} = \mathbf{R}^H \mathbf{U}^H \mathbf{U} \mathbf{R} = \mathbf{1}_M$, while they are correlated for oblique

⁵In theory, one could also rotate all components, but since the ultimate goal is a more interpretable solution in a lower dimensionality, one often transforms only the first components of the original PCA solution such that most of the variance is explained by them.

⁶In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [32, 42], but this is rarely done in practice anymore. The reason is that by rotating the unitary coefficient matrix, the orthogonality of the PCs is lost, but the orthogonality of the VCs is preserved, thus it is not useful to tackle effects like the Buell patterns. Additionally, the rotated VCs are rotated without taking into account the relative importance of the spatial patterns, which is reflected in the eigenvalues. As a consequence of this, the result of the rotation process becomes highly sensitive to the decision of how many components are to be rotated.

⁷Note that in the more general context of complex data, we would actually call this a unitary rotation. However, since factor analysis has been predominately applied to real-valued data, orthogonal and oblique rotations are historical terms. To avoid confusing naming conventions, I stick with the historical terms bearing in mind that we mean the more general case covering complex data.

rotation, for which $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{R}$. In both cases, however, the rotated loadings (or VCs) are not orthogonal anymore,

$$\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R = \mathbf{R}^H \tilde{\mathbf{V}}^H \tilde{\mathbf{V}} \mathbf{R} \stackrel{(1.16)}{=} \mathbf{R}^H \mathbf{A} \mathbf{R} \neq \mathbf{I}_M, \quad (1.20)$$

which is the key to obtain a more interpretable solution. Many simplicity criteria f have been proposed over the years [see e.g. 49], most of them are based on the idea that the rotated VCs should be more localized, less dense, and more physically interpretable. In practice, however, it was shown that the results are often not so sensitive to the actual choice of the simplicity criterion [49].

Varimax orthogonal rotation The most common orthogonal rotation technique is the varimax criterion [34], which seeks to simplify the VCs by maximizing the variance of the squared loadings,

$$f(\tilde{\mathbf{V}}_R) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^4 - \left(\sum_{j=1}^P b_{jm}^2 \right)^2 \right) \quad (1.21)$$

where b_{jm} are the elements of the rotated loadings $\tilde{v}_{R,jm}/h_j$ normalized by the square root of the *communalities* $h_j^2 = \sum_{m=1}^M \tilde{v}_{R,jm}^2$ of variable j . As a consequence of a varimax rotation, the elements of the rotated loadings matrix $\tilde{\mathbf{V}}_R$ are pushed towards either 0 or 1, which results in a sparse solution with mostly very large or very small values leading to a localization of the components. This is in particular useful when the underlying physical processes are expected to be localized in space, such as in the case of atmospheric or oceanic teleconnections.

Promax oblique rotation The most common oblique rotation technique is the promax criterion [26], which is a generalization of the varimax criterion to the oblique case. The promax criterion is based on the assumption that an orthogonal simple structure solution (e.g. varimax) is usually fairly close to an optimum oblique solution. As a result, the varimax-rotated loading elements $v_{R,jm}$ are used to construct a hypothetical oblique target matrix $\mathbf{Z}$ to be approximated by an oblique Procrustes transformation of the orthogonal varimax loadings matrix,

$$\mathbf{Z} = \tilde{\mathbf{V}}_R \mathbf{R} + \mathbf{E}, \quad (1.22)$$

1 where $\mathbf{T}$ is the $m \times m$ transformation matrix and $\mathbf{E}$ is a matrix of residuals. The solution of the
 2 Procrustes problem is given by the normal equation

$$3 \quad \mathbf{R} = (\tilde{\mathbf{V}}_R^T \tilde{\mathbf{V}}_R)^{-1} \mathbf{Z}^T \tilde{\mathbf{V}}_R. \quad (1.23)$$

4 Hendrickson & White [26] defined the target matrix $\mathbf{Z}$ in a way that increases the discrimination
 5 between high and low loadings, which is achieved by raising the loadings to the power of q
 6 but keeping the original sign,

$$7 \quad z_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (1.24)$$

8 where q is the power parameter that controls the degree of obliqueness and a_{jm} are the
 9 elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$ defined as

$$10 \quad \mathbf{A} = \tilde{\mathbf{A}} \mathbf{K}^{-1}, \quad (1.25)$$

$$11 \quad \tilde{\mathbf{A}} = \mathbf{H}^{-1} \tilde{\mathbf{V}}_R, \quad (1.26)$$

12 where $\mathbf{K}$ is a diagonal matrix with the diagonal elements being the maximum absolute values
 13 of the m -th column of $\tilde{\mathbf{A}}$ with diagonal elements $a_m = \max_m |A_{pm}|$ and $\mathbf{H}$ is a diagonal matrix
 14 with the diagonal elements being the square root of the communalities, h_j , of the j -th variable.

15 Both, orthogonal and oblique rotation techniques, in particular varimax and promax, have
 16 been used extensively since the 1980s in the literature to identify stationary modes of variability
 17 in climate data [e.g. 2, 18, 28, 31, 35, 39, 43]. Some of the most well-known climate oscillations,
 18 such as the North Atlantic Oscillation (NAO), has even been defined based on a Varimax-
 19 rotated pattern of loadings [2], while they also allowed to identify "ghost patterns", i.e. artificial
 20 patterns identified by PCA that are not physically meaningful [31, 38].

21 In practice, rotated modes have been associated with a number of additional benefits in the
 22 context of climate science. For instance, rotated modes are more robust against temporal
 23 and spatial subsampling [12] than unrotated modes, and rotation provides an efficient rem-
 24 edy against the statistical artifacts associated to the Buell patterns resulting in patterns that
 25 more realistically represent the underlying autocorrelation structure thus lead to increased
 26 interpretability [38, 49, 50].

Additionally, rotation techniques haven been successfully incorporated in other PCA extentions dealing with the non-linear and non-stationary nature of climate oscillations [8, 21, 27].

However, rotation techniques have also been subject to criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 32, pp. 270-274], which is why alternative approaches to enforce sparcity have been explored. In particular, Jolliffe *et al.* [33] introduced the Simplified Component Technique – LASSo (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from the context of regularized regression [53], which imposes a constraint on the ℓ_1 norm of the vector of loadings. An important feature of this method is that some of the loading coefficients are exactly zero.

SCotLASS suffers from the drawback that when the number of variables is much larger than the number of observations (i.e. $p \gg n$) then SCoTLASS only selects at most n non-zero elements in the loading vectors. This is of course not a desired property, as in climate science it's often the case that $p \gg n$ and that selecting only n variables is in general not very informative. Furthermore, since the solution of SCoTLASS cannot be obtained via SVD but requires a numerical approach using projected gradient approach [54], the computational demand is typically higher than for (rotated) PCA. Nevertheless, this publication has spurred the development of many other sparse PCA methods based on either the addition of a penalty or a constraint (usually ℓ_0 or ℓ_1) [see e.g. 16, 57, 58]. Yet, many of the sparse PCA formulations are complex to solve, and a considerable amount of work is of an algorithmic nature; proposed algorithms are often subject to local optima and without guaranteed convergence. Moreover, some of the procedures also fail in terms of memory, are very slow to compute [20] or have limited performance in modeling the data at hand [10].

Therefore, until today, rotated PCA remains one of the most commonly used techniques to obtain a sparse PCA solution in climate science, as it is computationally efficient and easy to implement. , and provides a

nicrie: clarify notation/terminology used for PCA eigenvectors, projections, scaled projections etc.

1.1.3 Hilbert PCA

PCA very efficiently identifies the principal axes along which the covariance among all variables is distributed, rotated PCA can help to relate these patterns to the underlying physical processes by providing a more sparse solution. However they all look at the current state of the vector field, they are *static* statistical methods. They ignore the time evolutions of the analyzed field. One important drawback of standard PCA is that is is a static method, i.e. it does not account for temporal dynamics or changes over time. In fact, any permutation of the

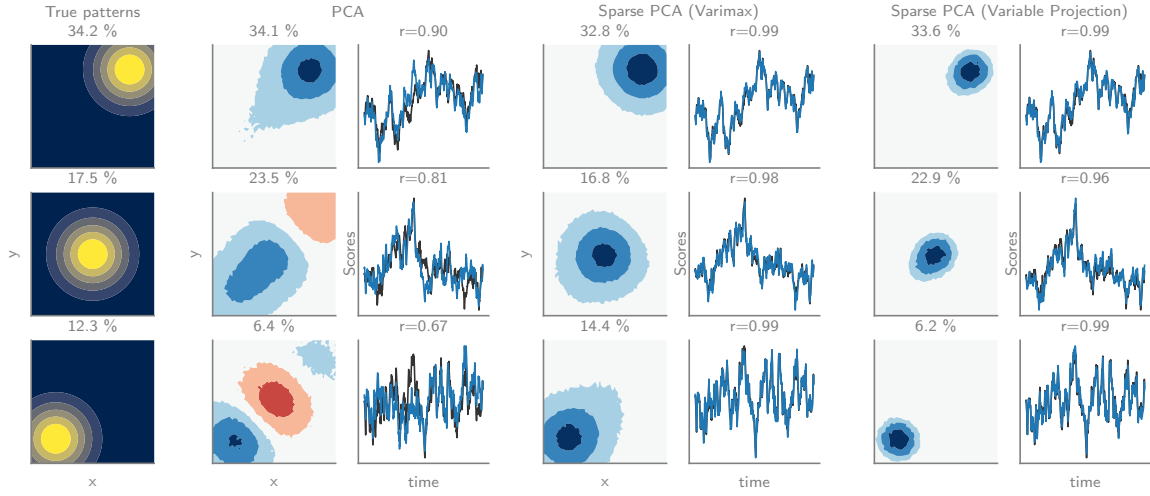


Fig. 1.1 Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

data points in time will lead to the same eigenvectors and eigenvalues, which can sometimes be a severe limitation in climate science where time evolution is a key aspect. Of particular interest in climate science are oscillatory patterns, such as the ENSO, QBO, MJO and BSISO, which all include spatial patterns that propagate in time. this aspect cannot be easily captured by standard PCA.

Hilbert Transform as a Momentum Operator

One way to overcome this limitation is to model not only the current 'state' of the climate but also its tendency or 'momentum', akin to the Hamiltonian principle in mechanics that the future of a system is described by a set of first-order differential equations for the location (state) and the momentum (tendency) [cf. 52, p. 353]. In classical mechanics, the most natural way to represent a propagating feature is by using the canonical coordinates known from the Hamiltonian formulation: each state variable is augmented by a momentum (or tendency) variable, and in the two-dimensional reference frame an oscillation is simply a point rotating in a circle. From a data point of view, this augmentation can be achieved by way of the Hilbert transform ($\mathcal{H}$). For a real time series $x(t)$, it is defined as

$$x'(t) = \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (1.27)$$

which is a reasonable measure of ‘momentum’ when variations in the data are confined to a relatively narrow frequency band [5, 17]. The p.v. denotes the Cauchy principal value of the integral, which ensures that the integral converges despite the singularity at $\tau = t$. In this form, the HT is seen as a convolution of $x(t)$ with $\frac{1}{\pi t}$, thus emphasising the local properties of $x(t)$.

However, more easily the HT can be understood in the frequency domain where it is related to the Fourier transform $\mathcal{F}$ via

$$\mathcal{F}\{y\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (1.28)$$

using the signum function sgn . The imaginary factor in Equation (1.28) simply means that the frequency components ω of x are phase-shifted by an angle of $\pi/2$, putting x' into quadrature to x . This frequency representation provides a more intuitive interpretation of the Hilbert transform. For instance, the HT of a sine wave is a cosine wave, and vice versa. In signal processing, the HT is known as a linear filter that doesn’t change the amplitude of the signal’s frequency components but alters their phase by 90 degrees.

The HT can be used to complexify a real-valued time series $x(t)$ by defining the complex time series $z(t) = x(t) + i\mathcal{H}\{x\}(t)$, which is called the *analytic signal* of $x(t)$, which provides a way to compute the instantaneous amplitude and phase of $x(t)$:

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (1.29)$$

Similar results can be derived for the discrete case [44, pp. 942-956] and can be readily extended to the multivariate case [e.g. 24, pp. 105-106], such that the analytical signal of a data matrix $\mathbf{X}$ is defined as

$$\mathbf{Z} = \mathbf{X} + i\mathcal{H}\{\mathbf{X}\}, \quad (1.30)$$

where $\mathcal{H}\{\mathbf{X}\}$ is the variable-wise applied Hilbert transform of $\mathbf{X}$.

Rasmusson *et al.* [47] used this idea of data augmentation to introduce the concept of Hilbert PCA⁸, which was later refined by Barnett [1] and Horel [28]. The idea is to apply PCA to the augmented data matrix $\mathbf{Z}$, which extends traditional real-valued PCA by incorporating phase

⁸Although Rasmusson *et al.* [47] used the term *Hilbert Singular Decomposition* in their original publication, we stick to *Hilbert EOF/PCA* as it is more commonly used in the literature [24, 32, 52]. Sometimes, *Complex EOF* analysis is also used [9].

information, which helps in capturing the direction and speed of wave-like patterns. Thus, it allows to identify and reconstruct propagating waves, whereas PCA only finds standing oscillations. The results obtained for PCA (Section 1.1) and rotated PCA (Section 1.1.2) still hold in this case, as we developed them for complex-valued data. Notably, the Hilbert PCA eigenvectors are complex, as are the principal component scores, and the eigenvectors are orthogonal in the complex sense, i.e. they form a unitary matrix. The real and imaginary part of the principal component scores are linked via the Hilbert transform, and the variance accounted for by each component is still represented by the real eigenvalues of the complex covariance matrix of Z .

The main advantage of Hilbert PCA over ordinary PCA is its ability to identify and reconstruct propagating waves, whereas PCA only finds standing oscillations. This is particularly useful in climate science, where many important climate oscillations are characterized by propagating waves, such as the MJO, ENSO, QBO, and BSISO. For instance, `cai_spatial-temporal_2001` applied Hilbert PCA to SST in the southern hemisphere and linked it to ENSO, while Bürger [9] used it to compute an index of the BSISO. However, only oscillatory patterns can only be robustly identified for long enough time series where which allow the resolution of at least one full oscillation [41]. Since the VCs of Hilbert PCA are orthogonal in a complex sense, they share the same limitations as PCA in terms of physical interpretability. Therefore, rotated Hilbert PCA has been proposed as a means to obtain a more interpretable solution [4, 28].

Finally, it is worth noting that there exists a closely related linear dimensionality reduction technique called canonical correlation analysis, or CCA (Hotelling, 1936; Bach and Jordan, 2002). Whereas PCA works with a single random variable, CCA considers two (or more) variables and tries to find a corresponding pair of linear subspaces that have high cross-correlation, so that each component within one of the subspaces is correlated with a single component from the other subspace. Its solution can be expressed in terms of a generalized eigenvector problem.

1.2 | Maximum Covariance Analysis

dating back to Tucker (1958), where it is known as inter-battery factor analysis; The method was popularised in atmospheric science by Bretherton et al. (1992) and Wallace et al. (1992) under the name singular value decomposition (SVD) analysis. the name maximum covariance analysis is used by von Storch and Zwiers (1999, Section 14.1.7)

difference to CCA: Maximum covariance analysis differs from CCA in two ways: covariance rather than correlation is maximized, and vectors of loadings are orthogonal instead of derived variates being uncorrelated. The rationale behind maximum covariance analysis is that it may be important to explain a large proportion of the variance in one set of variables using the

other set, and a pair of variates from CCA with large correlation need not necessarily have large variance

MCA reduces to PCA when both sets are the same, but there's an even deeper link to PCA: maximum covariance analysis can be viewed as two PCAs, followed by rotation to match up the results of the two analyses as closely as possible (Cherry, 1997).

Prediction via

$$\mathbf{P}_1 = \mathbf{X}_1 \mathbf{V}_1 \quad (1.31)$$

$$\mathbf{P}_2 = \mathbf{X}_2 \mathbf{V}_2 \quad \text{or} \quad \mathbf{P}_2 = \mathbf{P}_1 \mathbf{G} \quad \text{with} \quad g_k = \frac{s_k}{\text{varp}_{1,k}} \quad (1.32)$$

$$\mathbf{X}_2 = \mathbf{P}_2 \mathbf{V}_2^T = \mathbf{P}_1 \mathbf{G} \mathbf{V}_2^T = \mathbf{X}_1 \mathbf{V}_1 \mathbf{G} \mathbf{V}_2^T \quad (1.33)$$

Accuracy of reconstruction depends on

- The degree of covariance between the two fields.
- The cumulative squared covariance fraction represented by the leading modes. The more modes the better the reconstruction.
- The variance explained by the right singular vectors in the right phase space. I.e. the component of the right space that is related to the left space.

1.2.1 Rotated Maximum Covariance Analysis

1.2.2 Complex Maximum Covariance Analysis

CCA or MCA?

MCA focuses on covariance: Modes tend to be large where the variance is large. Danger that physical modes are confounded by large variance.

CCA focuses on correlation: Coupling is identified also if associated variance is low. Danger that physical modes are confounded by small (insignificant) variations (sampling problems)

1.3 | Canonical Correlation Analysis

1.3.1 Complex Canonical Correlation Analysis

1.4 | Neural Networks

1.4.1 Introduction to Neural Networks

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

1.4.2 Recurrent Neural Networks

Introduction to RNNs⁹: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM¹⁰: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.4.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

⁹Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

¹⁰Hochreiter & Schmidhuber 1997

Layerwise Relevance Propagation (LRP)

Concept of LRP¹¹: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables. simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP¹²: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

1.5 | Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

1.5.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

1.5.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

¹¹Bach et al. 2015

¹²Lundberg 2017

1.5.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1.5.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

1.6 | Comparative Perspective

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Proofs

Deriving PCA for the general M -dimensional projection space

Here we generalize the solution for the optimal linear projection for the case of an M -dimensional projection space. In Chapter 1.1, we have already shown that for $M = 1$, the variance of the projected data is maximized when $\mathbf{u}_1$ is the eigenvector of the covariance matrix $\mathbf{S}$ corresponding to the largest eigenvalue λ_1 .

Using proof by induction we will show that the optimal linear projection for an $(M - 1)$ dimensional subspace is given by the $M - 1$ eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$ corresponding to the $M - 1$ largest eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{M-1}$.

First, we need to find $\mathbf{u}_M$ such that the variance of the projection onto the M -dimensional subspace is maximized. The total variance for the M -dimensional projection space can be expressed as the sum of variances in each direction

$$\sum_{i=1}^M \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i$$

where $\mathbf{u}_M$ must be orthogonal to the previously chosen eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$:

$$\mathbf{u}_i^T \mathbf{u}_j = \delta_{ij} \quad \text{for } i, j = 1, 2, \dots, M.$$

Using the same methods as for the $M = 1$ case, we introduce a Lagrange multiplier λ_M and maximize

$$\mathbf{u}_M^T \mathbf{S} \mathbf{u}_M + \lambda_M (1 - \mathbf{u}_M^T \mathbf{u}_M).$$

Setting the derivative with respect to $\mathbf{u}_M$ to zero, we obtain

$$\mathbf{S} \mathbf{u}_M = \lambda_M \mathbf{u}_M$$

- ¹ showing that $\mathbf{u}_M$ is an eigenvector of $\mathbf{S}$ with eigenvalue λ_M .

